



Java Arrays

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Java Arrays

Arrays are used to store multiple values in a single variable, instead of declaring separate variables for each value.

To declare an array, define the variable type with **square brackets** `[ ]` :

```
String[] cars;
```

We have now declared a variable that holds an array of strings. To insert values to it, you can place the values in a comma-separated list, inside **curly braces** `{ }` :

```
String[] cars = {"Volvo", "BMW", "Ford", "Mazda"};
```

To create an array of integers, you could write:

```
int[] myNum = {10, 20, 30, 40};
```

Access the Elements of an Array



Example

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```
String[] cars = {"Volvo", "BMW", "Ford", "Mazda"};
System.out.println(cars[0]);
// Outputs Volvo
```

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Note: Array indexes start with 0: [0] is the first element. [1] is the second element, etc.

Think of an array as numbered boxes, where each box stores an element:

Index	Element
0	Volvo
1	BMW
2	Ford
3	Mazda

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Change an Array Element

To change the value of a specific element, refer to the index number:



Example

```
String[] cars = {"Volvo", "BMW", "Ford", "Mazda"};
cars[0] = "Opel";
System.out.println(cars[0]);
// Now outputs Opel instead of Volvo
```

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Array Length

To find out how many elements an array has, use the `length` property:

Example

```
String[] cars = {"Volvo", "BMW", "Ford", "Mazda"};
System.out.println(cars.length);
// Outputs 4
```

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The `new` Keyword

You can also create an array by specifying its size with `new`. This makes an empty array with space for a fixed number of elements, which you can fill later:



```
String[] cars = new String[4]; // size is 4

cars[0] = "Volvo";
cars[1] = "BMW";
cars[2] = "Ford";
cars[3] = "Mazda";

System.out.println(cars[0]); // Outputs Volvo
```

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However, if you already know the values, you don't need to write `new`. Both of these create the same array:

Example

```
// With new
String[] cars = new String[] {"Volvo", "BMW", "Ford", "Mazda"};

// Shortcut (most common)
String[] cars = {"Volvo", "BMW", "Ford", "Mazda"};
```

Note: You cannot write `new String[4] {"Volvo", "BMW", "Ford", "Mazda"}`.

In Java, when using `new`, you either:

- Use `new String[4]` to create an empty array with 4 slots, and then fill them later
- Or use `new String[] {"Volvo", "BMW", "Ford", "Mazda"}` (without specifying the number of elements) to create the array and assign values at the same time

Tip: The shortcut syntax is most often used when the values are known at the start. Use `new` with a size when you want to create an empty array and fill it later.

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Fill in the blanks to declare an integer array.

```
int [] numbers = {1, 2, 3, 4, 5};
```

char String double {}

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